Power BI Healthcare Analytics Project

Excel dimensional dataset → Power Query → Data model → DAX measures → Three-page interactive report

Platform: Power BI service (app.powerbi.com)

1. Dataset Overview

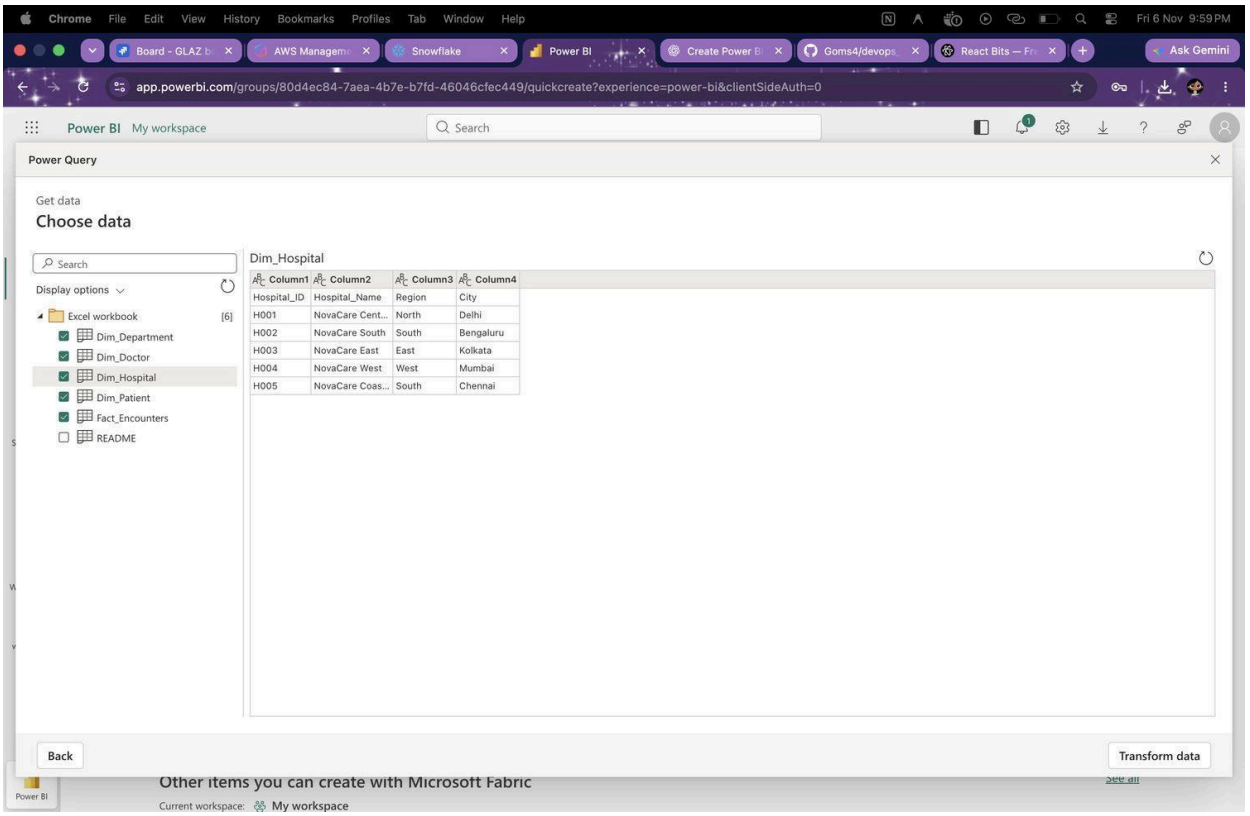
The Excel workbook contains six sheets. In the Choose data screen (Figure 1) the workbook node shows the count [6]: `Dim_Department`, `Dim_Doctor`, `Dim_Hospital`, `Dim_Patient`, `Fact_Encounters` and `README`. The first five were ticked for import; `README` was left unticked, which is consistent with it being a documentation sheet rather than analytical data.

Table	Type	Purpose and visible columns
Dim_Hospital	Dimension	Describes the five hospital sites. Columns: Hospital_ID, Hospital_Name, Region, City. Six rows including the header row: H001 NovaCare Central (North, Delhi), H002 NovaCare South (South, Bengaluru), H003 NovaCare East (East, Kolkata), H004 NovaCare West (West, Mumbai), H005 NovaCare Coastal (South, Chennai). The Hospital_Name for H001 and H005 is truncated in the preview ("NovaCare Cent…", "NovaCare Coas…"); the full names appear in the report charts (Figure 5, Figure 14).
Dim_Department	Dimension	Describes eight departments. Columns: Department_ID, Department_Name, Department_Type, Visit_Share (decimal). Departments D01 to D08 are Cardiology, Orthopedics, Neurology, General Medicine, Emergency, Pediatrics, Oncology and Gastroenterology; types shown are Medical, Surgical, Critical Care and Specialty (Figure 6).
Dim_Doctor	Dimension	Describes 20 doctors (DR001 to DR020). Columns: Doctor_ID, Doctor_Name, Department_ID, Experience_Years (whole number). Each doctor is assigned to one of D01 to D08 (Figure 7).
Dim_Patient	Dimension	Describes patients. Columns: Patient_ID, Age (whole number), Gender, Patient_Segment (Adult, Pediatric or Senior). IDs run from P0001 upward; the preview shows more than 99 rows (Figure 8).
Fact_Encounters	Fact	One row per hospital encounter. Starts with 21 columns and grows to 24 after the custom columns. Visible columns include Encounter_ID, Encounter_Date, Patient_ID, Doctor_ID, Department_ID, Hospital_ID, Admission_Type, visit type, Insurance_Type, Room_Charges, Total_Bill, Insurance_Coverage, Patient_Payment, Satisfaction_Score, Outcome and Readmission_30_Days (Figure 2 to Figure 4, Figure 9).

Encounter IDs follow the pattern E00001, E00002 and so on. The values seen in `Admission_Type` are Referral, Scheduled and Walk-in; in `Outcome` they are Recovered, Stable, Referred and Follow-up Required; and in `Readmission_30_Days` the preview rows show No. The `Visit_Share` column in `Dim_Department` adds up to 1.00 across the eight departments (0.18 + 0.14 + 0.12 + 0.20 + 0.16 + 0.08 + 0.07 + 0.05), so it appears to represent each department's expected share of visits.

2. Data Import Using Power Query

Figure 1: Selecting the dimension and fact tables from the Excel workbook in Power Query



Explanation:

The Power Query dialog (Get data > Choose data) is open inside the Power BI service. The left pane lists the Excel workbook with six items. `Dim_Department`, `Dim_Doctor`, `Dim_Hospital`, `Dim_Patient` and `Fact_Encounters` are ticked, and `README` is not. Choosing only the data sheets keeps the model free of non-analytical content.

3. Data Preparation and Transformation

After Transform data was selected, the Power Query editor opened with five queries (the "Queries [5]" pane), one per table. The screenshots show the Applied steps below for each query. Steps named Source and Navigation are created automatically when a workbook table is picked.

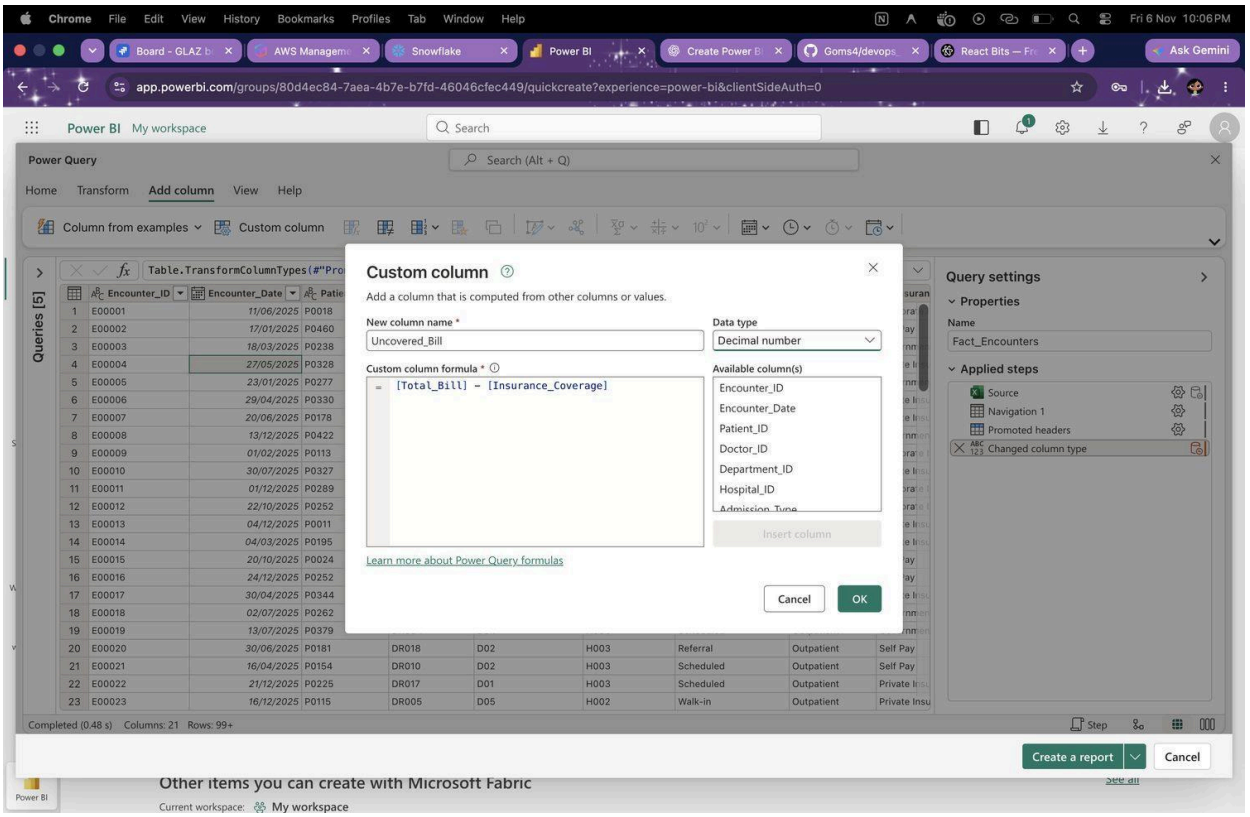
Query	Applied steps visible	Observation
Fact_Encounters	Source, Navigation 1, Promoted headers, Changed column type, Added custom, Added custom 1, Added custom 2	Three calculated columns added; 21 columns grow to 24 (Figure 2 to Figure 4)
Dim_Hospital	Source, Navigation 1, Changed column type	No Promoted headers step is visible, and columns are still Column1 to Column4 (Figure 5)
Dim_Department	Source, Navigation 1, Promoted headers, Changed column type	Visit_Share typed as decimal (Figure 6)
Dim_Doctor	Source, Navigation 1, Promoted headers, Changed column type	Experience_Years typed as whole number (Figure 7)
Dim_Patient	Source, Navigation 1, Promoted headers, Changed column type	Age typed as whole number (Int64) (Figure 8)

3.1 Promoting headers and changing column types

3.2 Calculated columns in Fact_Encounters

Three custom columns were added from the Add column tab using the Custom column dialog. They are calculated in Power Query, so they are stored as ordinary columns in the loaded fact table.

Figure 2: Adding the Uncovered_Bill custom column in Power Query

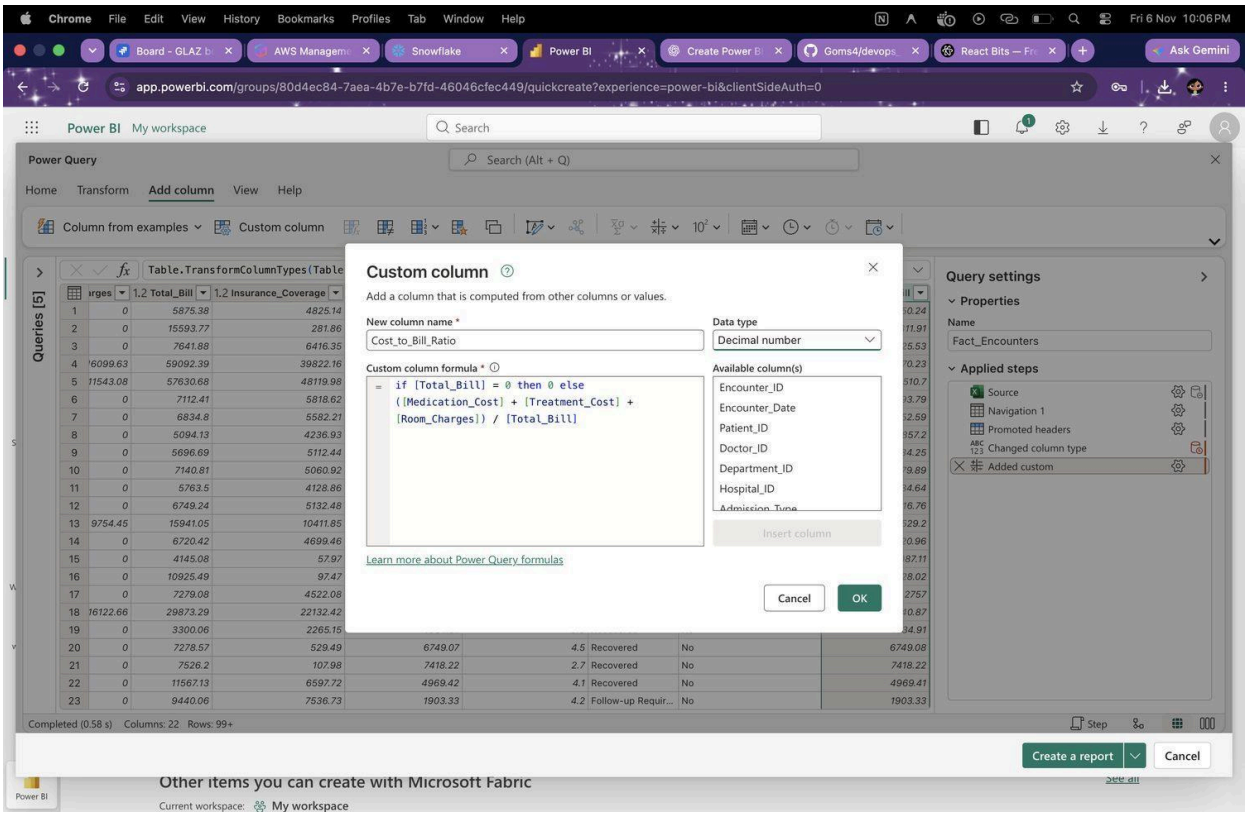


Explanation:

The Add column tab is active and the Custom column dialog is open over `Fact_Encounters`. The new column is named `Uncovered_Bill`, its data type is Decimal number, and the formula is:

```
[Total_Bill] - [Insurance_Coverage]
```

Figure 3: Adding the Cost_to_Bill_Ratio custom column with a conditional formula

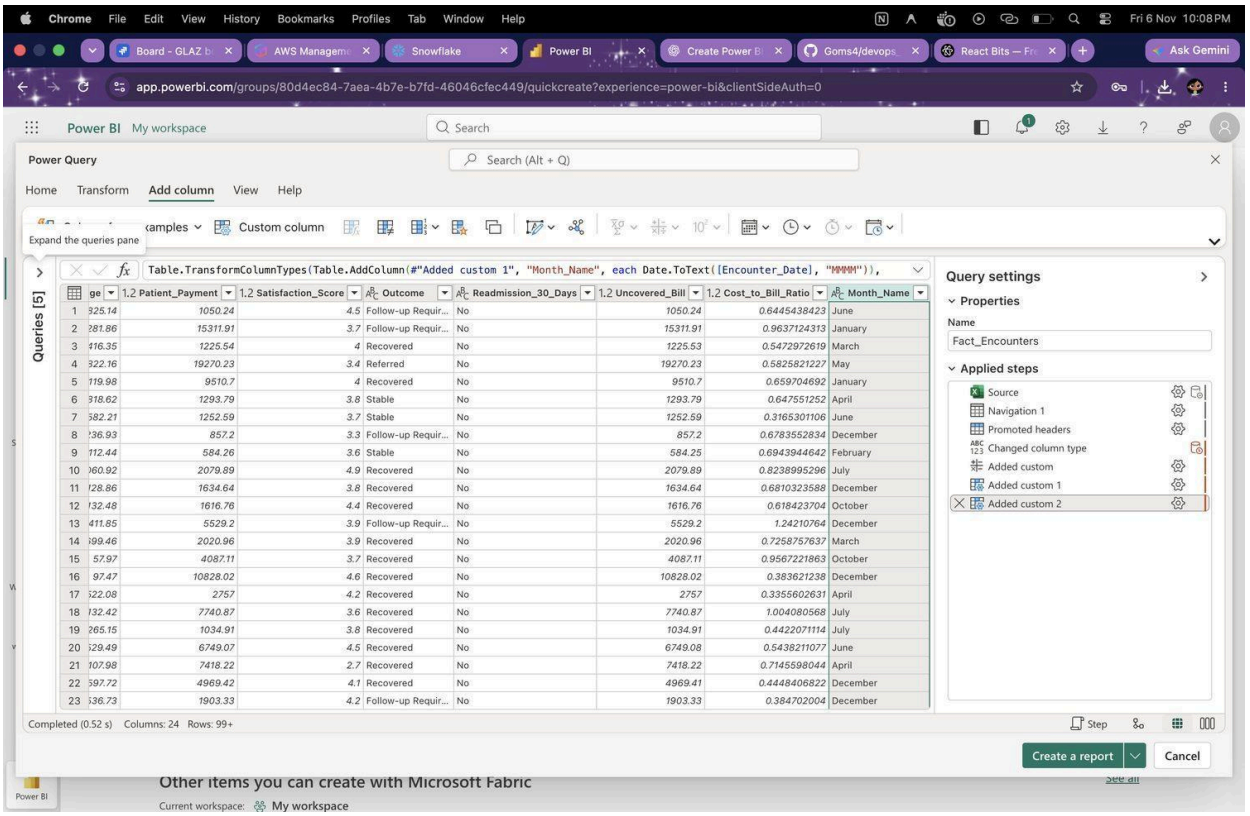


Explanation:

A second Custom column dialog is shown. The column `Cost_to_Bill_Ratio` is typed as Decimal number. The formula is:

```
if [Total_Bill] = 0 then 0 else ([Medication_Cost] + [Treatment_Cost] +
[Room_Charges]) / [Total_Bill]
```

Figure 4: Adding the Month_Name column and reviewing the extended fact table



Explanation:

The formula bar shows the third calculated column: `Table.AddColumn(#"Added custom 1", "Month_Name", each Date.ToText([Encounter_Date], "MMMM"))`. It converts each encounter date into the full month name (June, January, March and so on). Applied steps now end with Added custom, Added custom 1 and Added custom 2, and the table has 24 columns.

7.3 Reviewing the dimension tables

Each dimension was opened in turn in the query list to confirm its structure before modelling.

Figure 5: Dim_Hospital query with its Applied steps

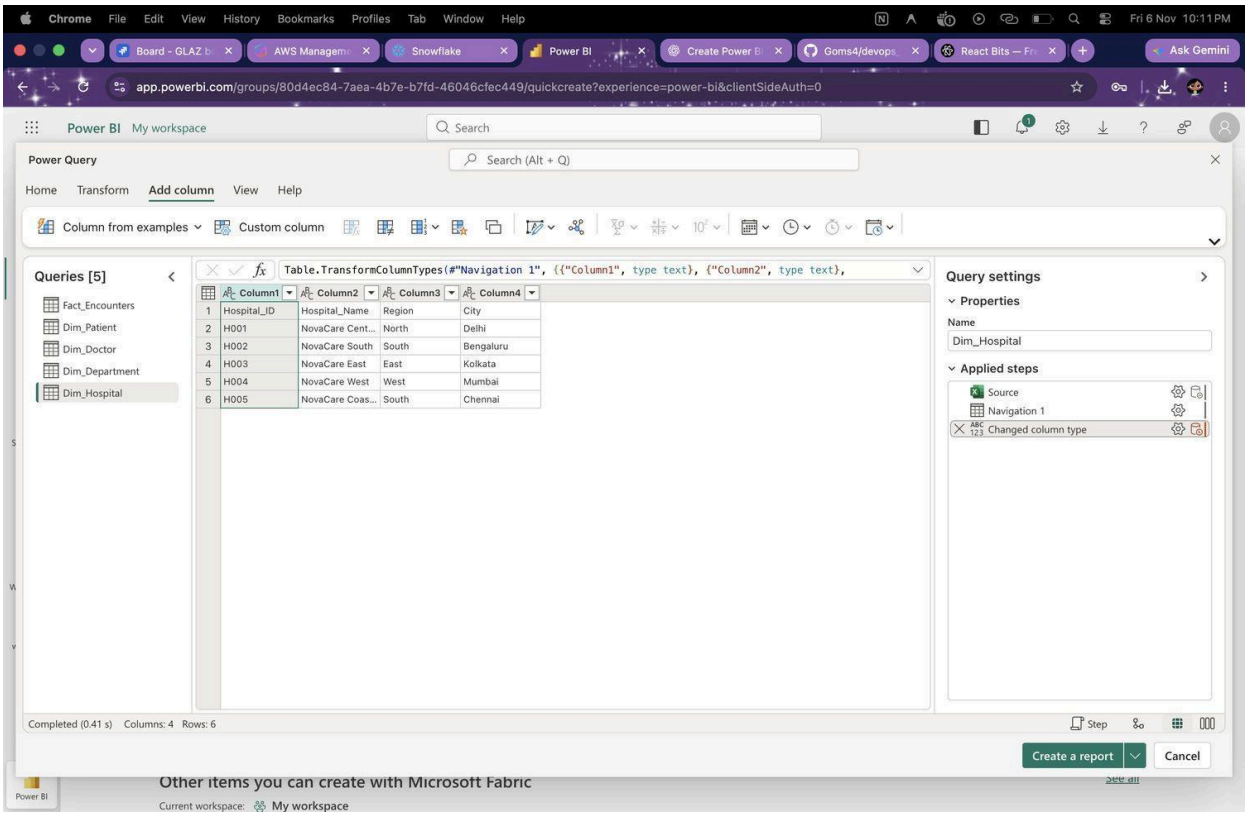


Figure 6: Dim_Department query

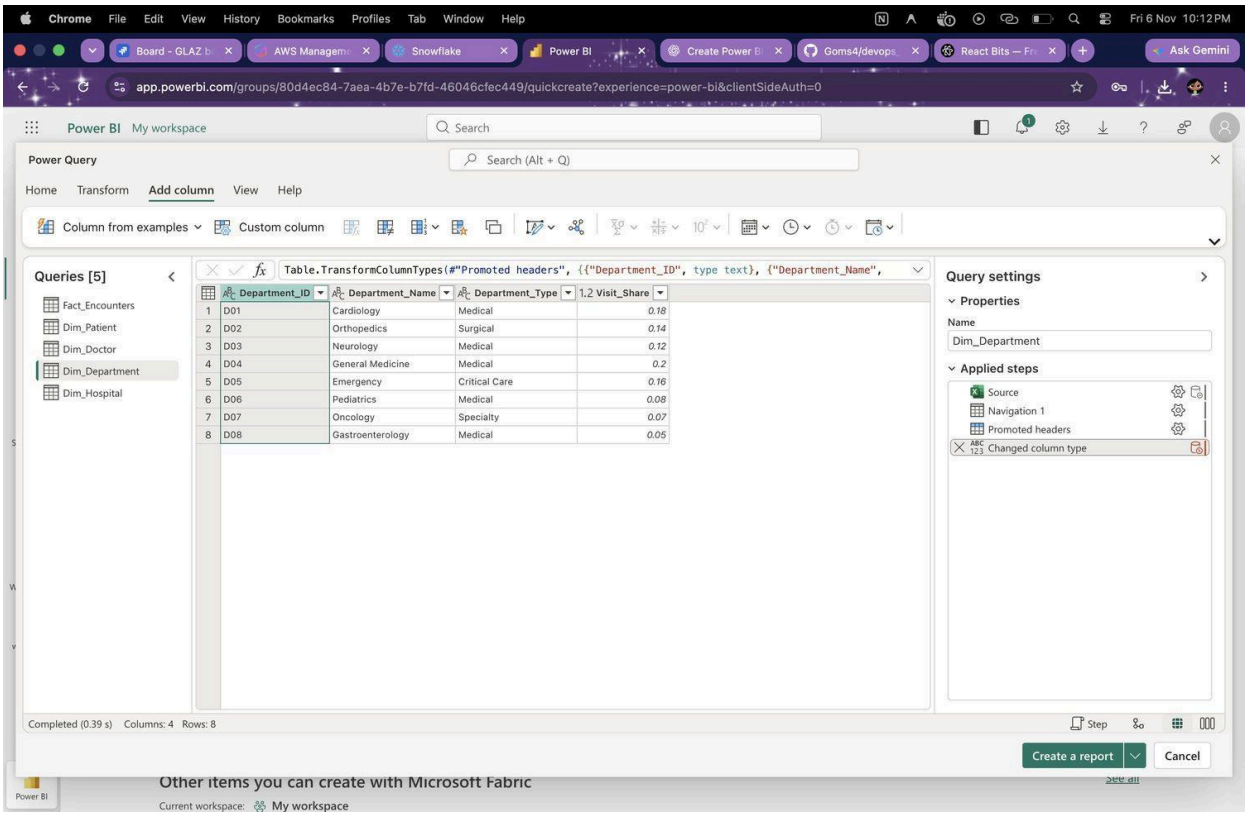


Figure 7: Dim_Doctor query

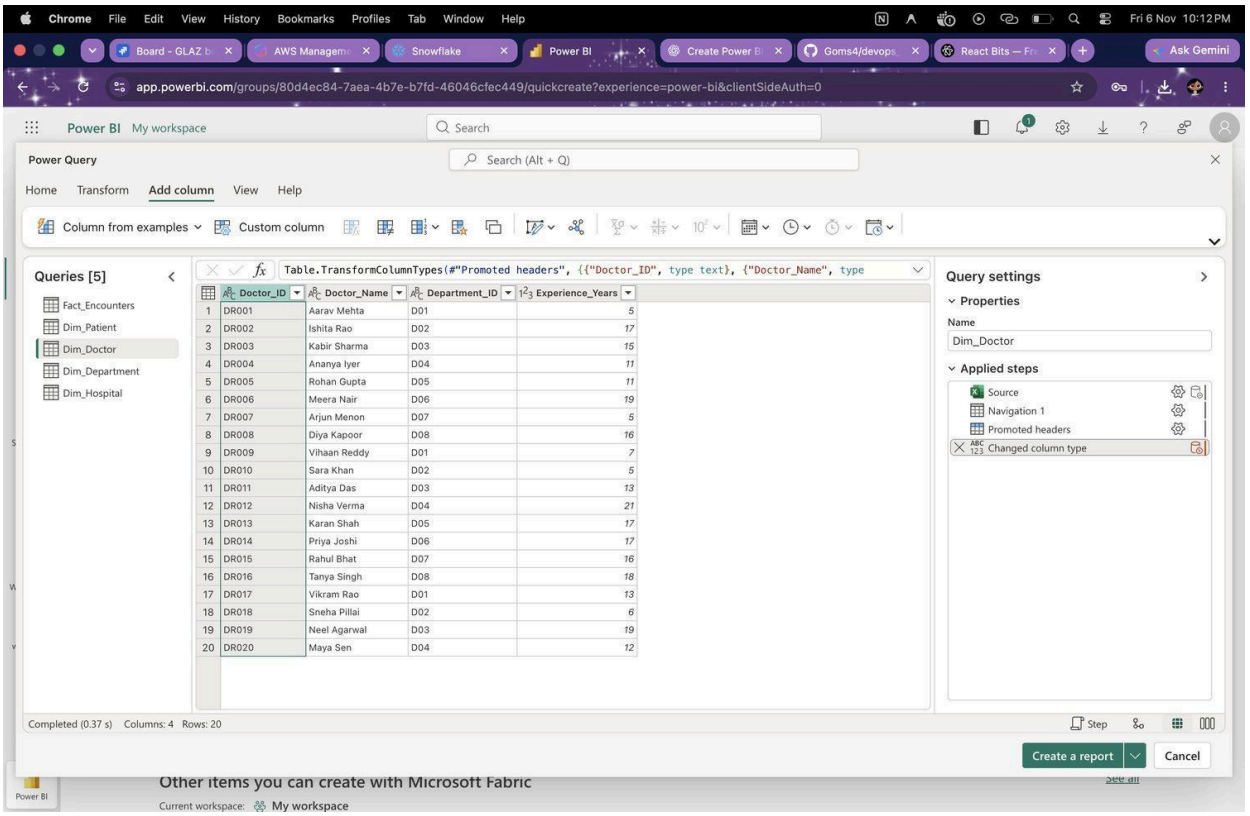
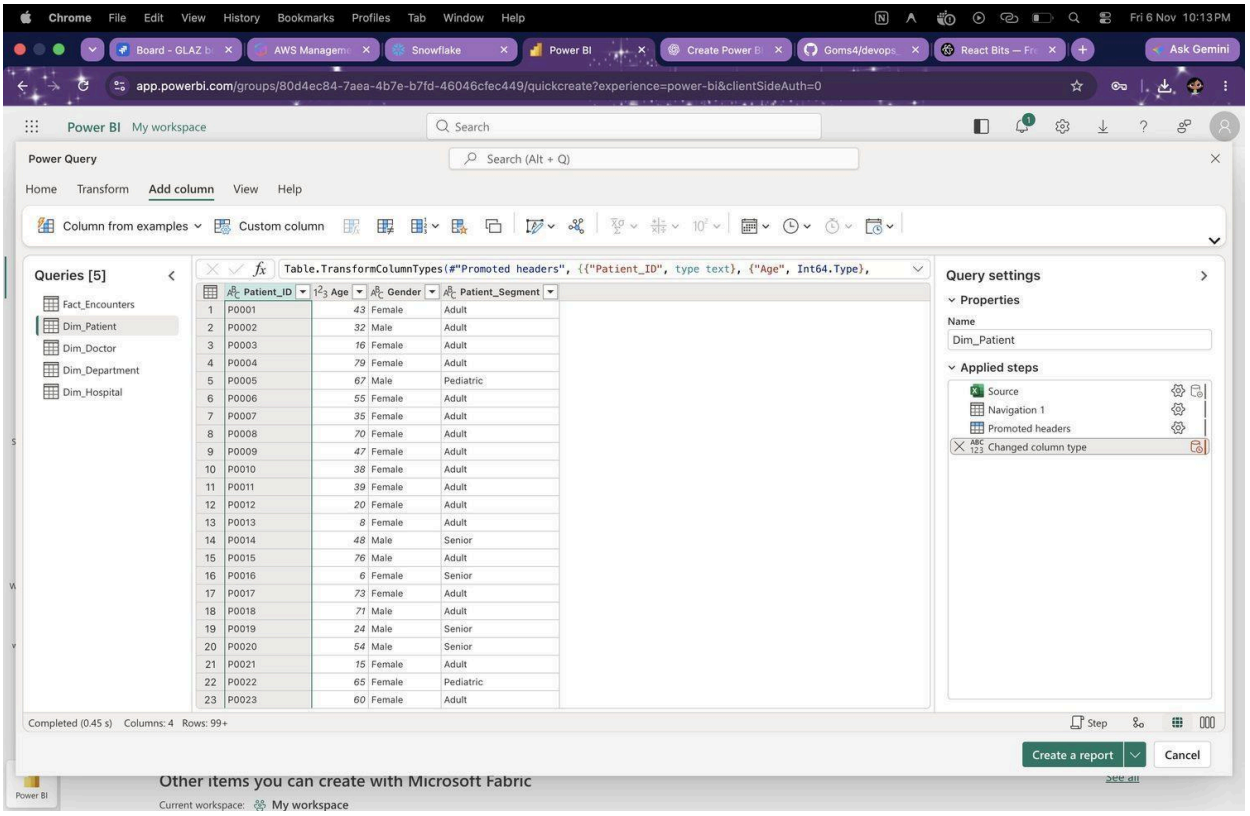


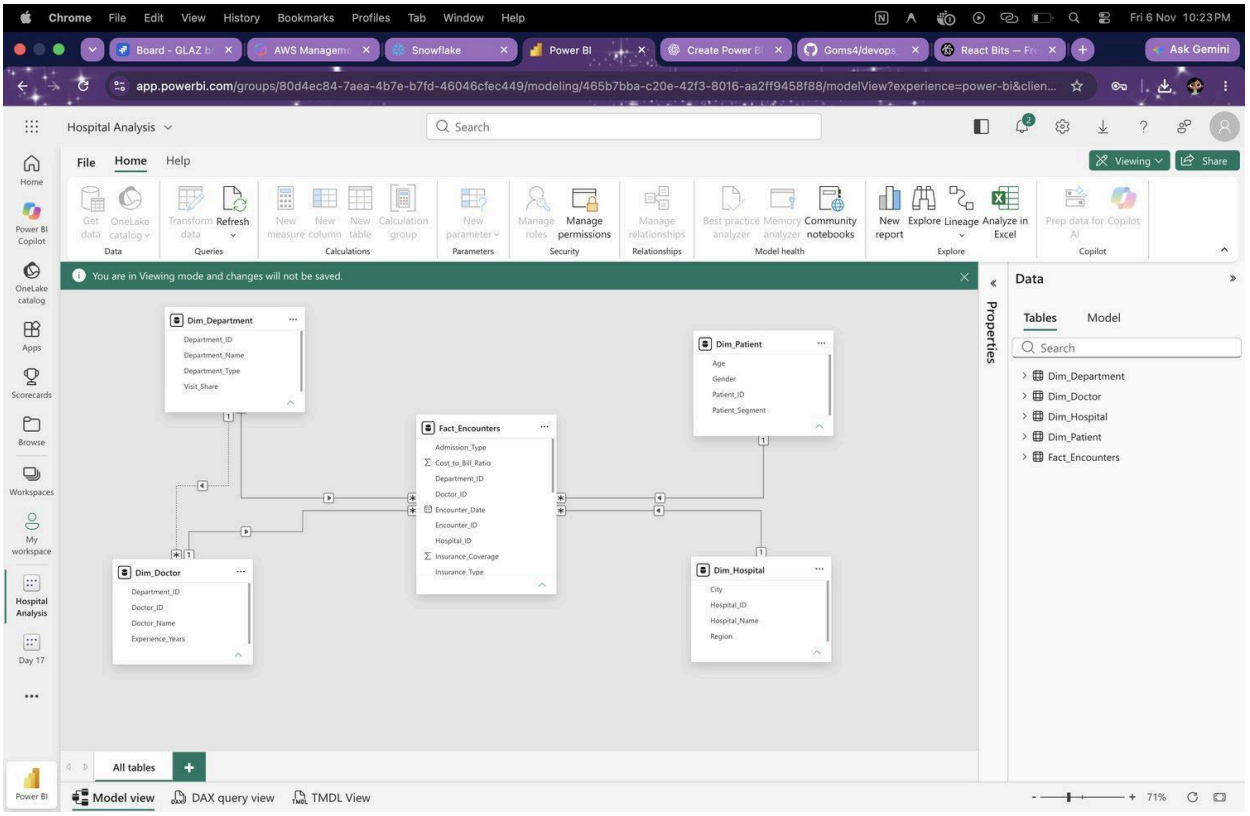
Figure 8: Dim_Patient query



4. Data Modelling

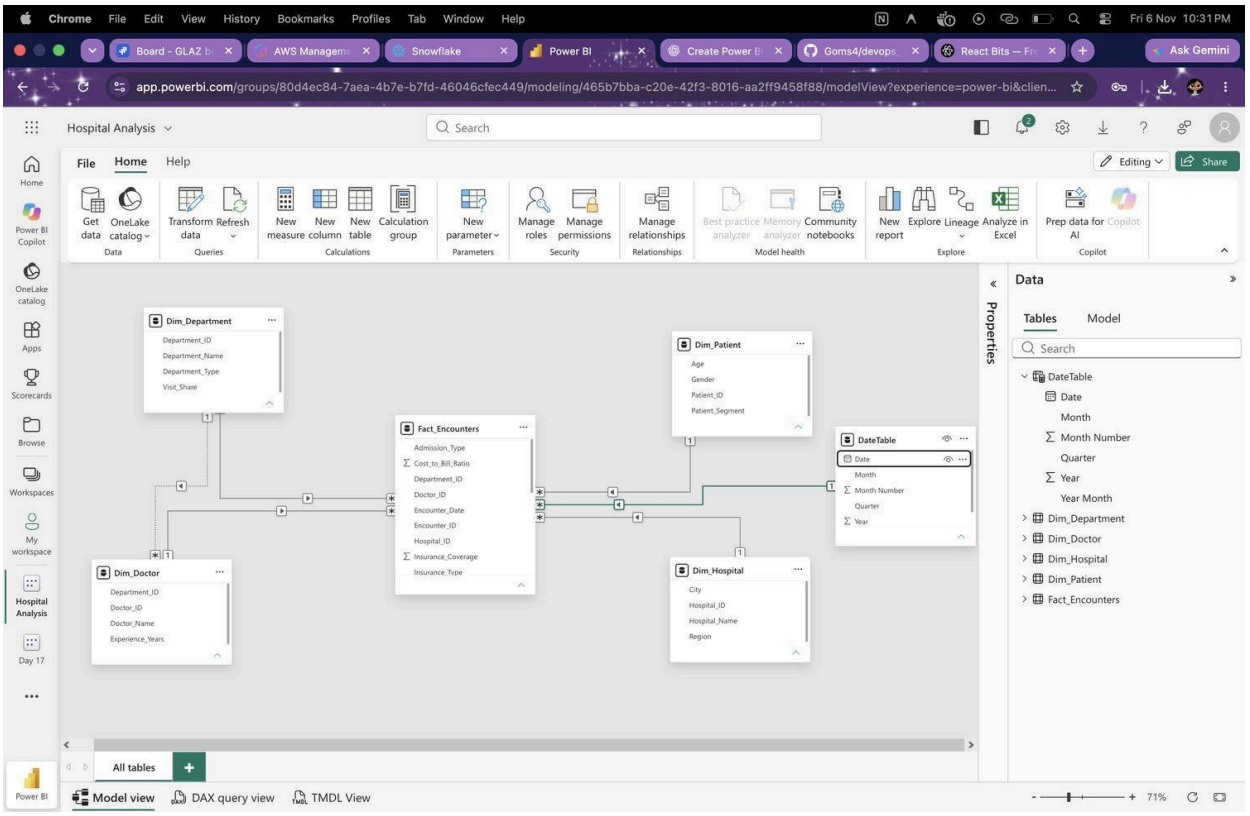
Once the queries were created, the tables were loaded into a semantic model named **Hospital Analysis**. The Model view (a tab at the bottom of the screen next to DAX query view and TMDL View) shows the tables as cards joined by relationship lines.

Figure 9: Model view of the Hospital Analysis semantic model



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Figure 10: DateTable added to the model and related to Fact_Encounters



Explanation:

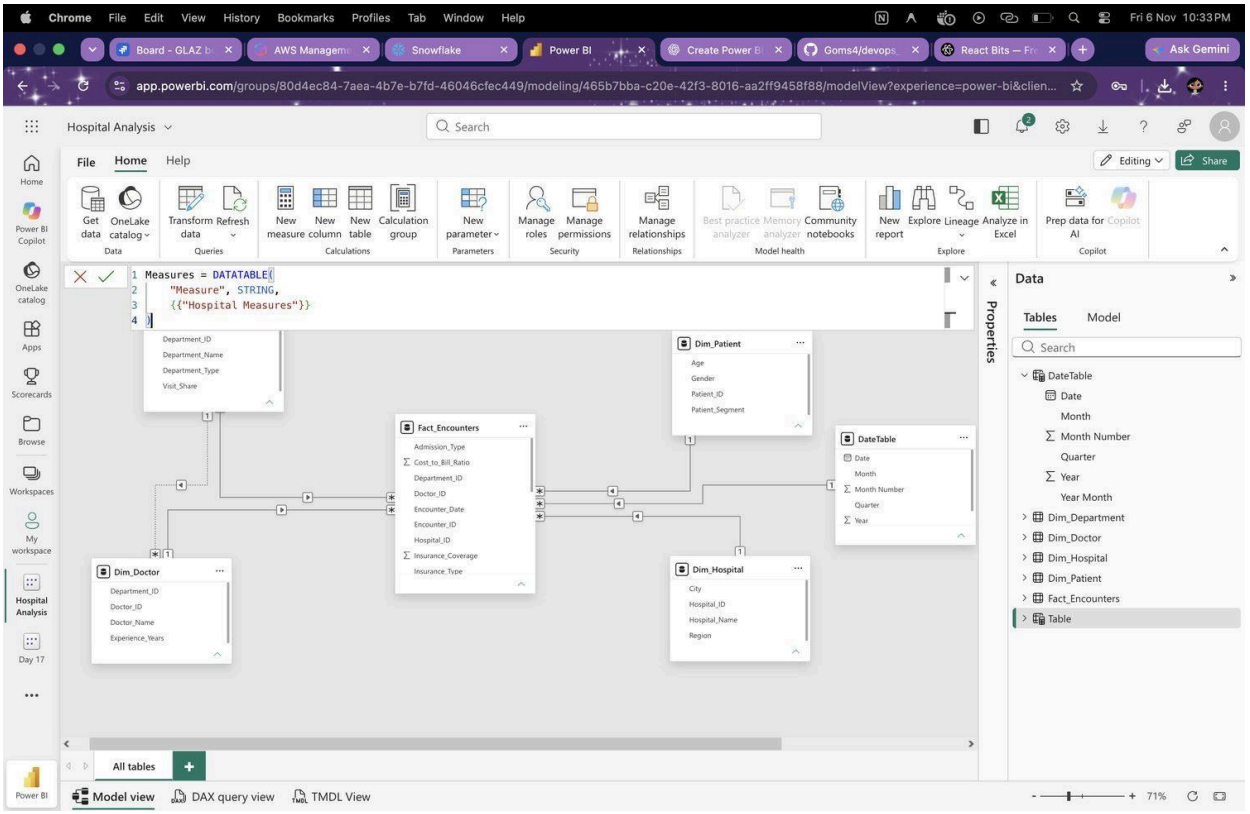
The session is now in Editing mode. A new table called `DateTable` appears on the right with the columns Date, Month, Month Number, Quarter, Year and Year Month .

5. DAX Measures and Calculations

Measures are calculations written in DAX that are evaluated at report time against the current filters. The screenshots show how a dedicated measure table was created and the names of the measures it contains.

5.1 Creating a table to hold the measures

Figure 11: Creating a measure container table with DATATABLE



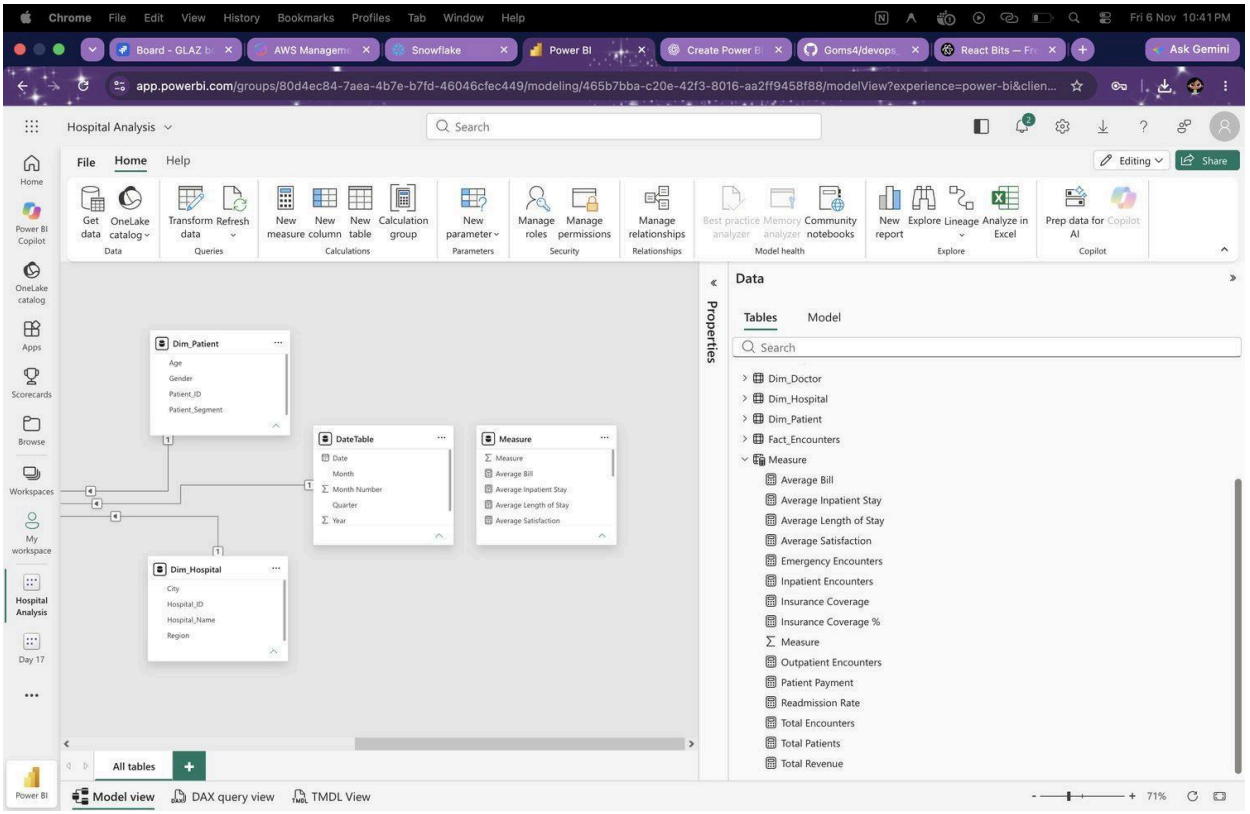
Explanation:

The DAX formula bar is open in Model view with the following expression:

```
Measures = DATATABLE(
    "Measure", STRING,
    {{"Hospital Measures"}}
)
```

5.2 Measures created

Figure 12: Measure table listing the hospital measures



Explanation:

The visible measures are listed in the table below.

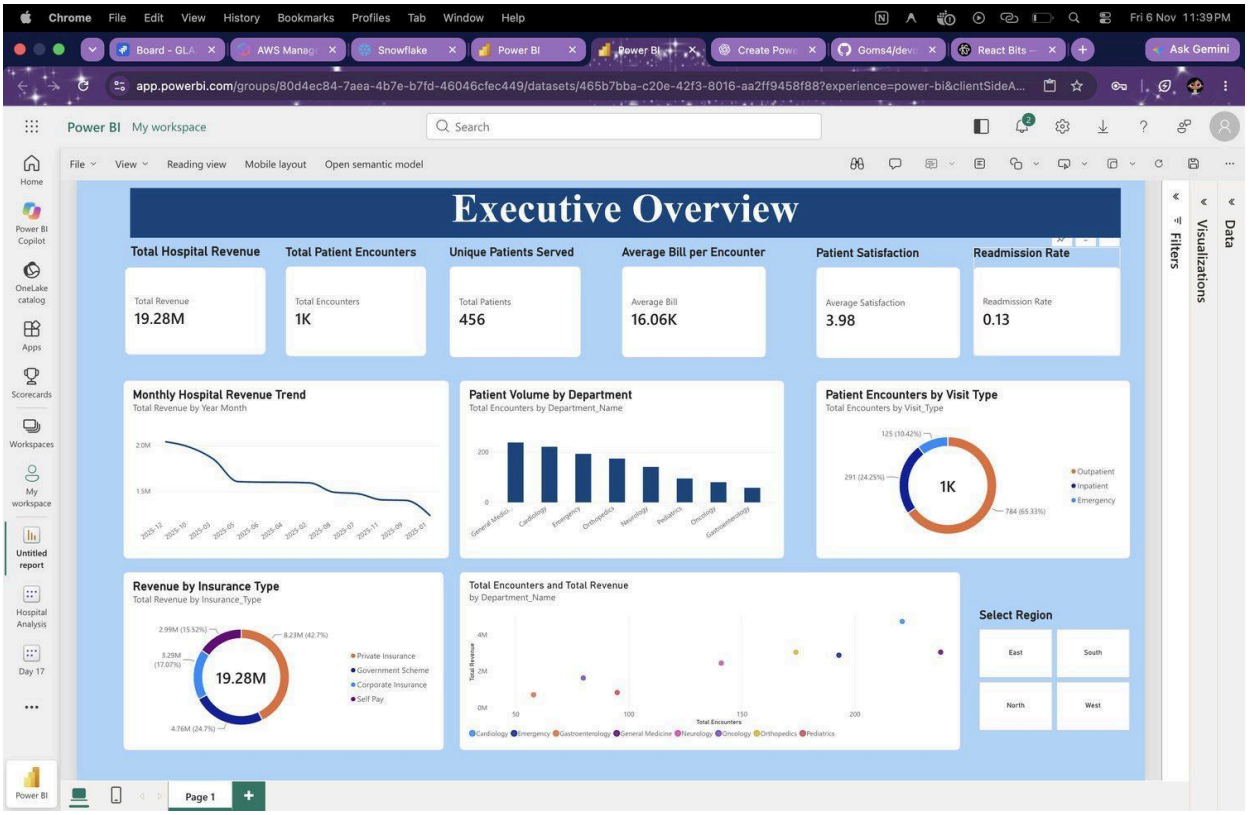
Measure name	Where it is used in the report	Business meaning (from the name and report values)
Total Revenue	KPI cards, revenue charts (all three pages)	Total billed revenue. 19.28M across all data. Equals Patient Payment + Insurance Coverage (7.23M + 12.05M) (Figure 15).
Total Encounters	KPI card, department, visit type and scatter visuals	Number of encounters. Shown as 1K (1,200 when the visit type split is added up).
Total Patients	KPI card (Unique Patients Served)	Count of distinct patients, 456 overall.
Average Bill	KPI cards	Average bill per encounter, 16.06K overall.
Average Satisfaction	KPI cards	Mean patient satisfaction score, 3.98 overall.
Readmission Rate	KPI cards	Share of readmissions within 30 days, shown as 0.13.
Average Inpatient Stay	Page 2 card and department chart	Average inpatient length of stay, 6.78 overall.
Emergency Encounters	Page 2 card	Count of emergency-type encounters, 125.
Patient Payment	Page 3 card and stacked chart	Amount paid by patients, 7.23M.
Insurance Coverage	Page 3 card and stacked chart	Amount covered by insurance, 12.05M.

Measure name	Where it is used in the report	Business meaning (from the name and report values)
Average Length of Stay, Inpatient Encounters, Outpatient Encounters, Insurance Coverage %	Listed in the measure table	Names indicate length of stay, encounter counts by visit type and the insured share of billing.

6. Report and Dashboard Development

6.1 Page 1: Executive Overview

Figure 13: Executive Overview page with no region selected



Explanation:

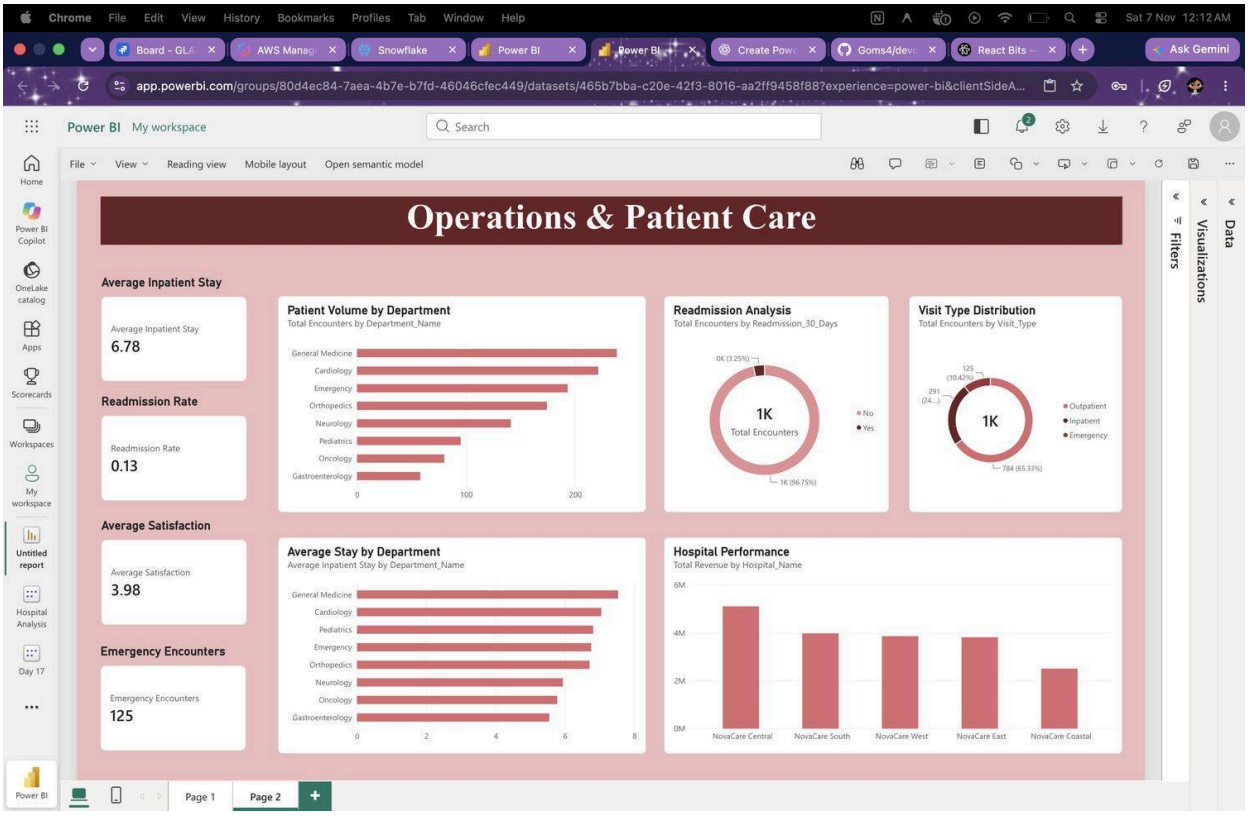
The page "Executive Overview" and six KPI cards: Total Hospital Revenue (19.28M), Total Patient Encounters (1K), Unique Patients Served (456), Average Bill per Encounter (16.06K), Patient Satisfaction (3.98) and Readmission Rate (0.13). Below the cards are five visuals and a region slicer. These are the unfiltered, hospital-wide figures, as no region tile is highlighted.

Visual	Type	Fields	Business question
KPI cards (6)	Card	Total Revenue, Total Encounters, Total Patients, Average Bill, Average Satisfaction, Readmission Rate	What is the overall size and quality of the operation?
Monthly Hospital Revenue Trend	Line chart	Total Revenue by Year Month	How does revenue change over time?
Patient Volume by Department	Column chart	Total Encounters by Department_Name	Which departments see the most patients?

Visual	Type	Fields	Business question
Patient Encounters by Visit Type	Donut chart	Total Encounters by Visit_Type	How are encounters split between outpatient, inpatient and emergency care?
Revenue by Insurance Type	Donut chart	Total Revenue by Insurance_Type	Which payer types bring in the most revenue?
Total Encounters and Total Revenue by Department	Scatter chart	X: Total Encounters, Y: Total Revenue, legend: Department_Name	Do busier departments also earn more?
Select Region	Slicer (tiles)	Region (East, South, North, West)	How do the results change for one region?

6.2 Page 2: Operations & Patient Care

Figure 14: Operations & Patient Care page



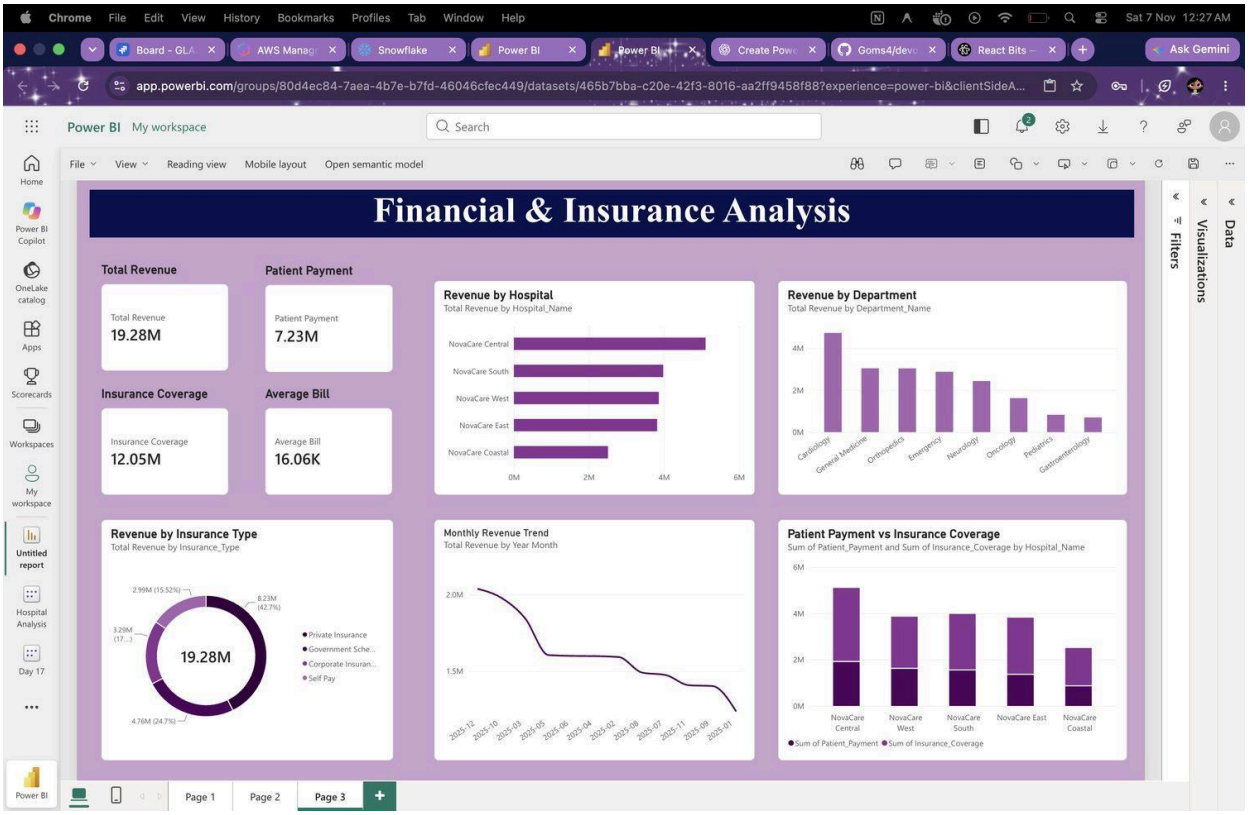
Explanation:

The page has four cards: Average Inpatient Stay (6.78), Readmission Rate (0.13), Average Satisfaction (3.98) and Emergency Encounters (125). Four charts follow:

Patient Volume by Department is a horizontal bar chart of Total Encounters, led by General Medicine and Cardiology and ending with Gastroenterology. Readmission Analysis is a donut of Total Encounters by Readmission_30_Days: "No" is 96.75% and "Yes" is 3.25%. Visit Type Distribution repeats the visit type split (Outpatient 784 or 65.33%, Inpatient 291 or 24.25%, Emergency 125 or 10.42%). Average Stay by Department shows average inpatient stay by department, which ranges up to about 8 days for General Medicine and about 5 to 6 for Gastroenterology and Oncology. Hospital Performance is a column chart of Total Revenue by Hospital_Name: NovaCare Central is highest (around 5M), then South, West, East and Coastal (about 2.5M).

6.3 Page 3: Financial & Insurance Analysis

Figure 15: Financial & Insurance Analysis page



Explanation:

The page has four cards: Total Revenue (19.28M), Patient Payment (7.23M), Insurance Coverage (12.05M) and Average Bill (16.06K). The charts are: Revenue by Hospital (bar chart, Total Revenue by Hospital_Name); Revenue by Department (column chart, led by Cardiology at roughly 4.5M and with Pediatrics and Gastroenterology lowest); Revenue by Insurance Type (donut with the same 8.23M, 4.76M, 3.29M and 2.99M split as Page 1); Monthly Revenue Trend (line chart by Year Month, with the same axis ordering issue noted above); and Patient Payment vs Insurance Coverage (stacked column chart by hospital, with the sum of Patient_Payment and the sum of Insurance_Coverage stacked for each hospital).

7. Filters and Interactivity

The only slicer visible is the **Select Region** tile slicer on Page 1, with the options East, South, North and West. Selecting a tile filters every visual on the page: the region value filters `Dim_Hospital`, which filters `Fact_Encounters` through the Hospital_ID relationship, and every card and chart recalculates. Four screenshots capture the page with each region selected in turn.

Figure 16: Executive Overview filtered to the East region

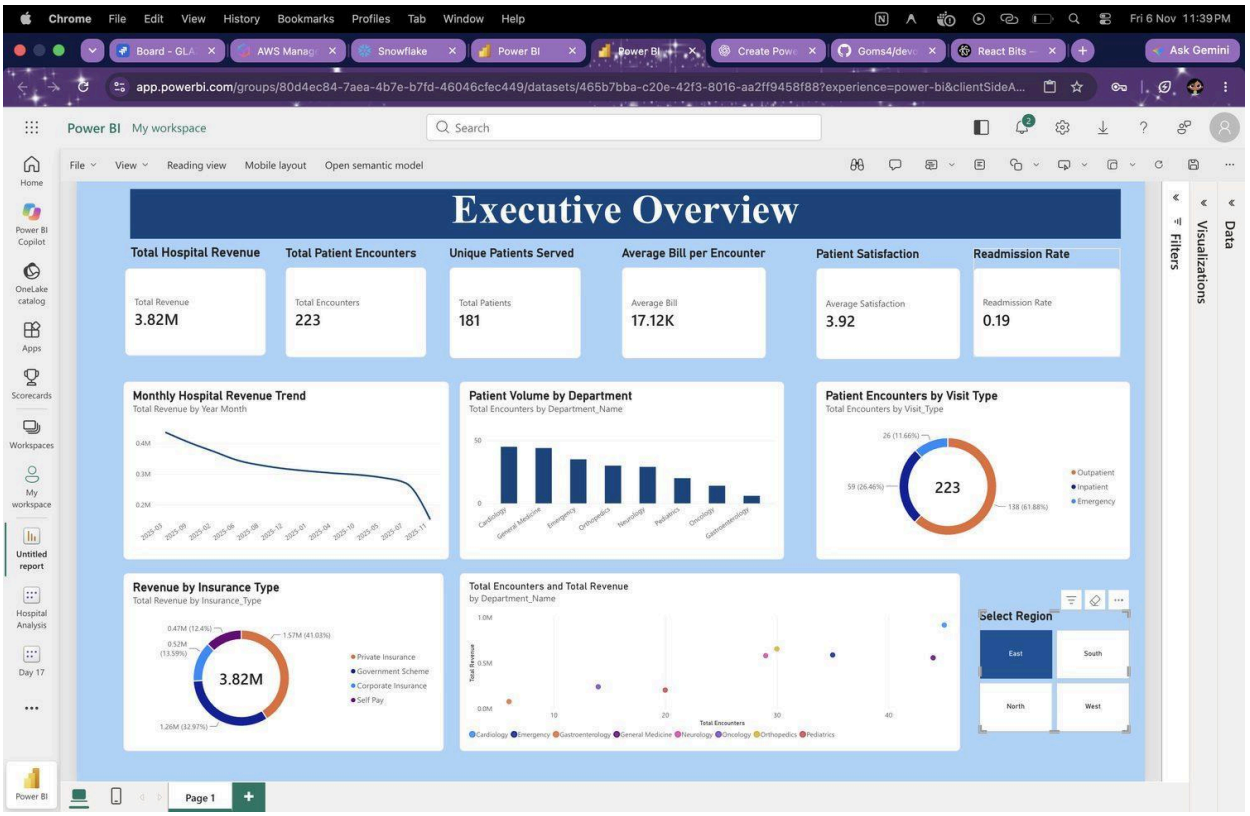


Figure 17: Executive Overview filtered to the South region

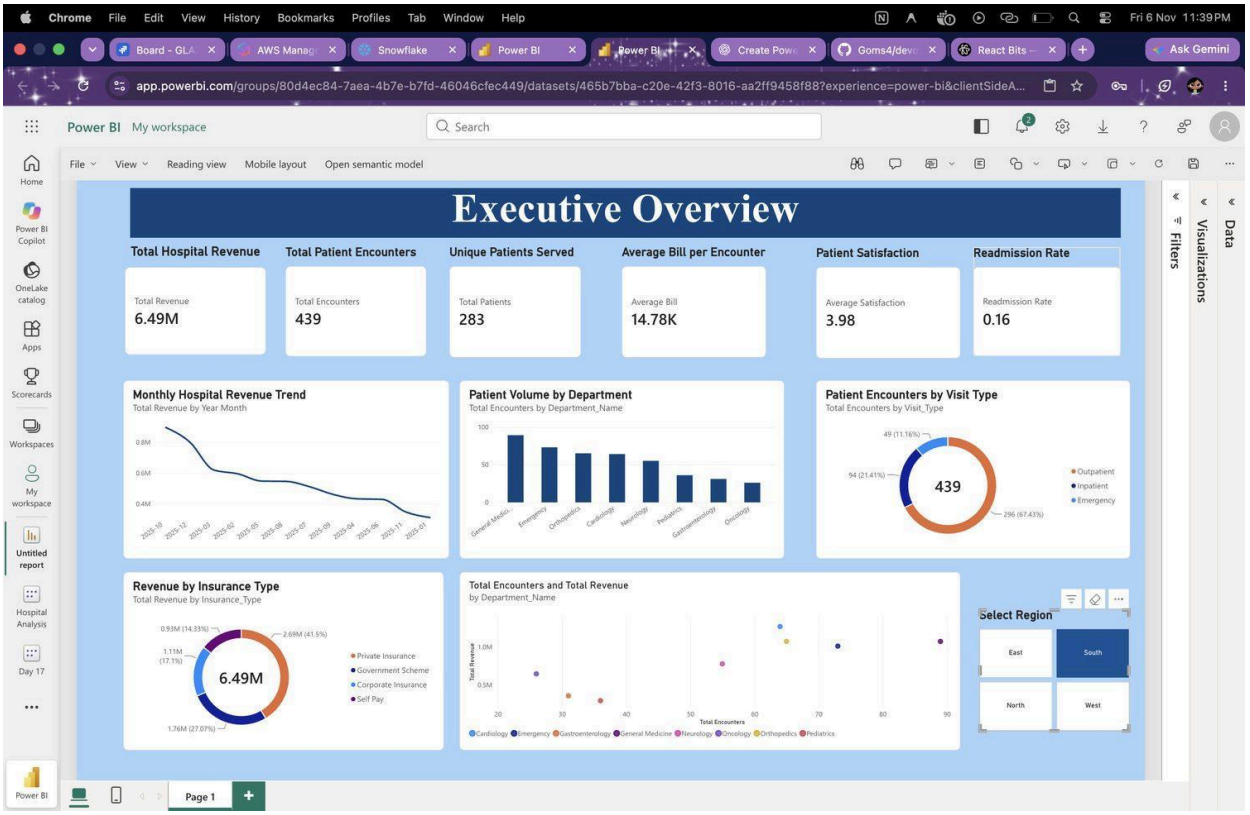


Figure 18: Executive Overview filtered to the North region

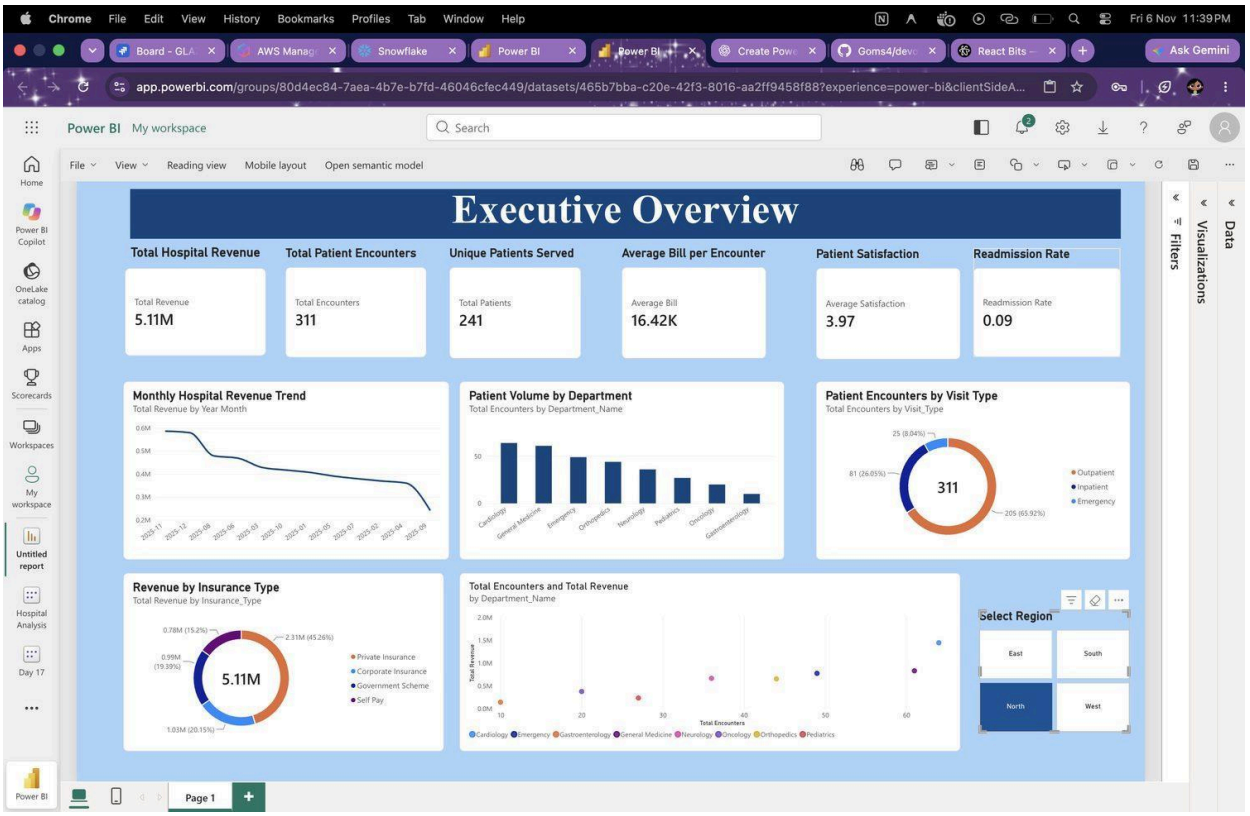
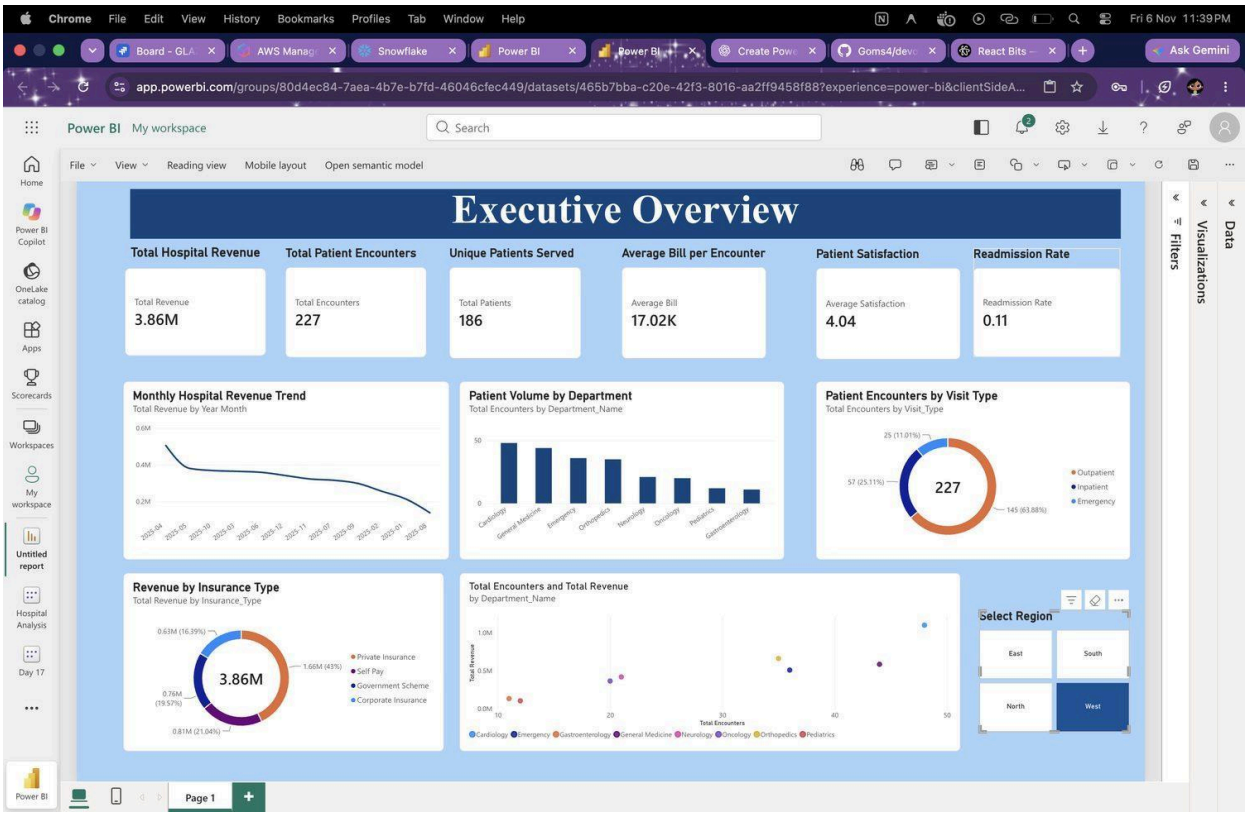


Figure 19: Executive Overview filtered to the West region



8. Key Insights

Region	Revenue / Encounters / Avg bill / Satisfaction / Readmission
East	3.82M / 223 / 17.12K / 3.92 / 0.19
South	6.49M / 439 / 14.78K / 3.98 / 0.16
North	5.11M / 311 / 16.42K / 3.97 / 0.09
West	3.86M / 227 / 17.02K / 4.04 / 0.11
All regions	19.28M / 1,200 / 16.06K / 3.98 / 0.13

- **Outpatient care dominates activity.** Outpatient encounters are 784 of 1,200 (65.33%), inpatient 291 (24.25%) and emergency 125 (10.42%) (Figure 13, Figure 14).
- **Insurance is the main source of revenue.** Insurance coverage is 12.05M of 19.28M (about 62.5%) and patient payment is 7.23M. By insurance type, Private Insurance contributes the most (8.23M, 42.7%), followed by Government Scheme (4.76M), Corporate Insurance (3.29M) and Self Pay (2.99M) (Figure 13, Figure 15).
- **South is the largest region by volume and revenue** (6.49M, 439 encounters) but has the lowest average bill (14.78K). East and West have the highest average bills (17.12K and 17.02K) with the lowest volumes.
- **Readmission rate differs by region.** East is highest at 0.19 and North lowest at 0.09, against 0.13 overall. East also has the lowest satisfaction score (3.92), while West has the highest (4.04). Satisfaction is otherwise very close across regions (3.92 to 4.04).
- **Hospital and department concentration.** NovaCare Central has the highest revenue of the five hospitals and NovaCare Coastal the lowest (Figure 14, Figure 15). Cardiology leads on revenue, and General Medicine and Cardiology lead on encounter volume; Gastroenterology and Pediatrics are at the low end on revenue.
- **General Medicine has the longest average inpatient stay** among departments on Page 2, with an overall average of 6.78.
- **Time trend cannot be concluded.** Because the monthly line charts are not sorted chronologically, the screenshots do not support a statement about whether revenue is rising or falling.

9. End-to-End Workflow

Stage	What was done
Excel dataset	Workbook with five data sheets and a README sheet (Figure 1).
Power BI Get data	Excel source selected in the Power BI service quick-create flow.
Power Query	Tables previewed and loaded as five queries (Figure 1, Figure 5 to Figure 8).
Data cleaning / transformation	Headers promoted, data types set, and the columns Uncovered_Bill, Cost_to_Bill_Ratio and Month_Name added (Figure 2 to Figure 4).
Data model	Semantic model "Hospital Analysis" created from the loaded tables (Figure 9).
Relationships	One-to-many links from the four dimensions and from DateTable to Fact_Encounters (Figure 9, Figure 10).
DAX measures	Measure table created and populated with 14 measures (Figure 11, Figure 12).

Stage	What was done
Visualisations	Three report pages of KPI cards and charts (Figure 13 to Figure 15).
Interactive dashboard	Region slicer filtering all Page 1 visuals (Figure 16 to Figure 19).
Healthcare insights	Regional, departmental, visit type and insurance comparisons (Section 12).